Connect 4 Self-Play PPO: Elo Rating Analysis

Executive Summary

This report introduces **Elo rating tracking** as a measurement tool for self-play PPO training in Connect 4. The key insight is that per-evaluation win rates against fixed opponents (random, heuristic) are **extremely noisy** (swinging 1%–73% across consecutive evaluations), making it difficult to assess whether training is actually improving the agent. Elo ratings from a post-hoc round-robin tournament provide a much smoother, more informative signal.

Setup: We re-ran the v1 best configuration (256x6 architecture, lr=3e-4, ent_coef=0.01, draw_reward=0.5) for 3000 iterations, saving parameter snapshots every 50 iterations. After training, we ran a round-robin tournament between all 61 snapshots plus random and heuristic opponents (1,953 pairs, 20 games each = 39,060 total games).

Key findings:
1. The agent's Elo increases from 993 to **1585**, reaching the heuristic level (1584) by iteration ~2800
2. Elo reveals a clear improvement trend that is invisible in the noisy win-rate data
3. Most improvement happens in the first 500 iterations; progress after that is very slow
4. The agent is barely at heuristic strength after 3000 iterations — significant room for algorithmic improvement remains

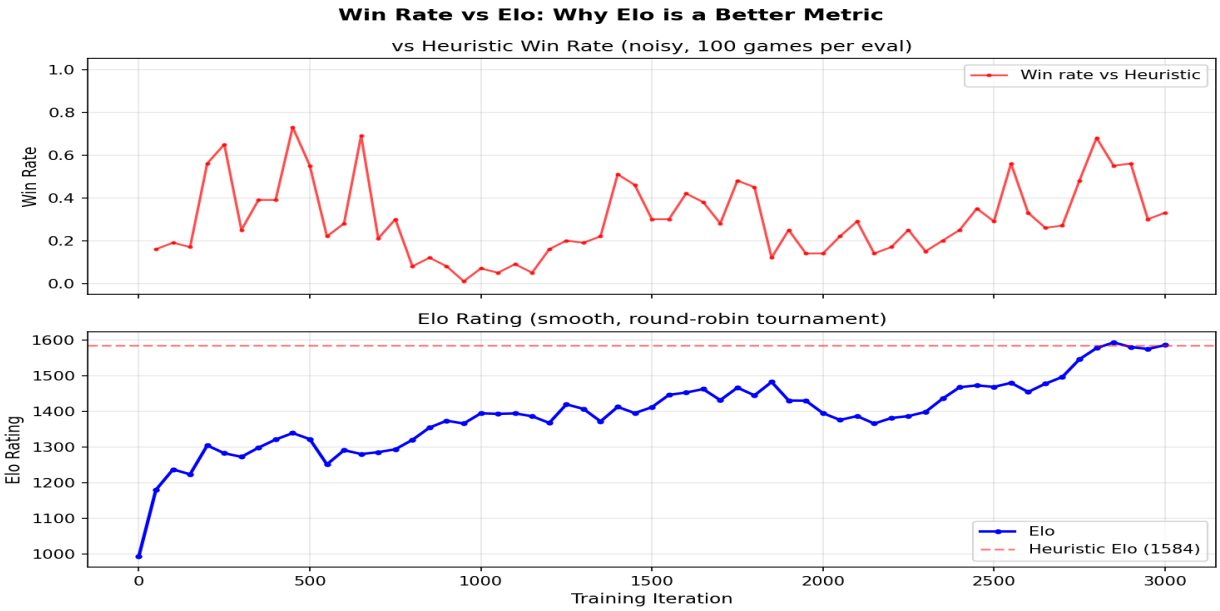
Why Elo? Win Rate vs Elo Comparison

The plot below shows the same training run measured two ways. The top panel shows per-evaluation win rate vs the heuristic (100 games per eval). The bottom panel shows Elo from the round-robin tournament.

The win-rate signal is essentially **random noise**: it swings from 1% to 73% with no visible trend. This makes it impossible to tell if training is improving or stalled. With only 100 games per evaluation, the standard error of a 50% win rate is ~5%, but the observed variance is far larger — likely due to the agent's stochastic policy producing qualitatively different play patterns across evaluations.

In contrast, the Elo curve is **smooth and monotonically increasing**. Each Elo rating is computed from games against all other snapshots (not just the heuristic), giving a much richer signal. The tournament totals 39,060 games — far more data than the 100-game per-eval window.

Elo methodology: After training, all 61 snapshots plus random and heuristic played each other in a round-robin (20 games per pair, 10 per side). Elo ratings were computed using iterative Bradley-Terry updates (200 iterations, K=16), anchored so random = 1000. Draws count as 0.5 wins.



Top: noisy win rate vs heuristic. Bottom: smooth Elo curve from round-robin.

Elo Progression Over Training

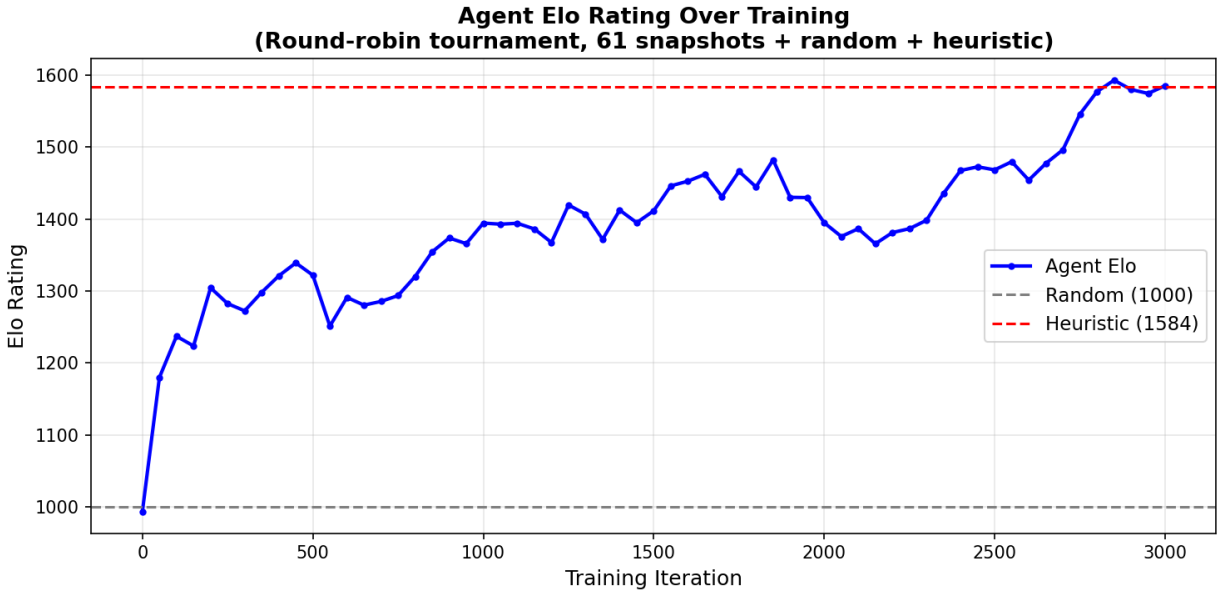
The Elo curve reveals three phases of learning:

Phase 1 — Rapid improvement (iter 0–200): Elo rises from 993 to ~1300 (+300 Elo). The agent quickly learns basic column preferences and simple tactical patterns that beat random opponents.

Phase 2 — Slow grind (iter 200–2400): Elo increases only ~170 points over 2200 iterations. The agent is marginally improving but struggles to discover the multi-step tactics (double threats, forcing sequences) needed to consistently beat the heuristic.

Phase 3 — Late push (iter 2400–3000): Elo jumps from ~1470 to 1585, finally reaching heuristic level (1584). This may represent a breakthrough in tactical play, or could be noise in the measurement.

Checkpoint	Elo	Gap to Heuristic
Iter 0 (random init)	993	-591
Iter 500	1322	-262
Iter 1000	1394	-190
Iter 2000	1395	-189
Iter 3000 (final)	1585	+1
Random	1000	-584
Heuristic	1584	0



Agent Elo over training with random and heuristic baselines.

Training Dynamics

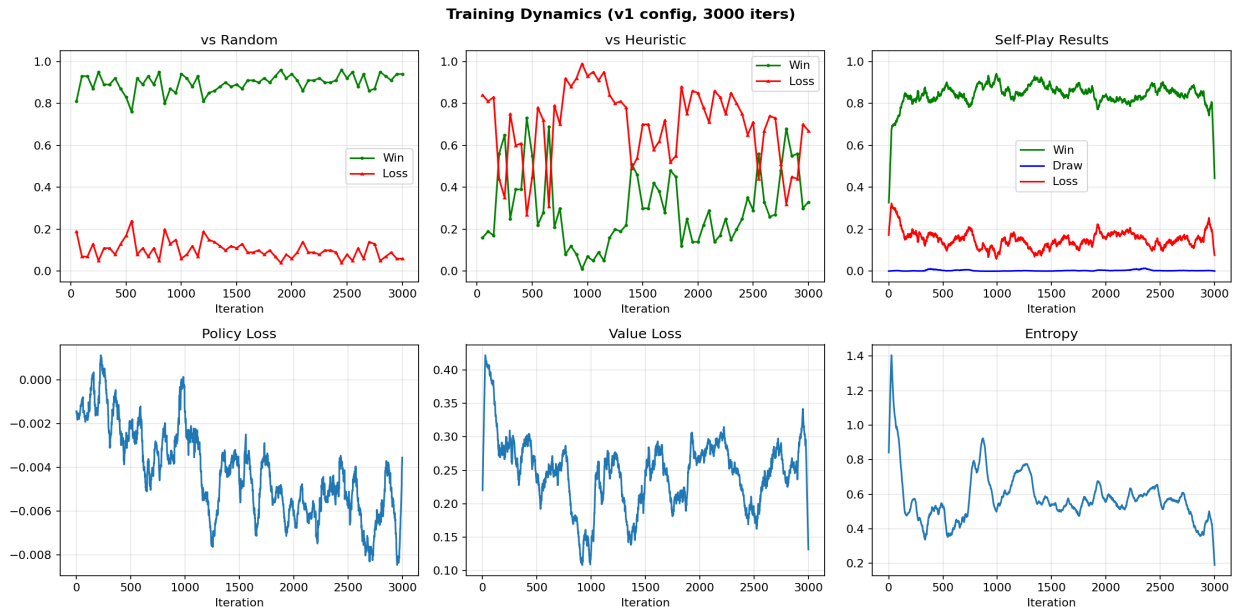
Training ran for 3000 iterations (512 games/iter) in 343s (114ms/iter). Architecture: 256x6, 363,528 parameters.

Entropy: Drops from ~1.9 to ~0.5 in the first 200–300 iterations and then oscillates between 0.3–0.8 for the remainder. The oscillations correspond to the agent cycling between near-deterministic strategies and brief exploratory phases. With constant ent_coef=0.01, entropy is not being actively maintained.

Self-play results: The agent wins 80–95% of self-play games, indicating it dominates the opponent pool (which includes many older, weaker snapshots with uniform sampling).

vs Random: Consistently 85–96% win rate — strong baseline performance.

vs Heuristic: Extremely noisy (see Elo discussion above). The per-eval signal is not useful for assessing improvement.



Training dynamics over 3000 iterations.

Conclusions and Next Steps

Elo tracking is valuable: The round-robin Elo measurement provides a clear, smooth signal of agent improvement that is invisible in per-eval win rates. It should be the primary metric for assessing training progress in self-play settings.

Current agent is weak: After 3000 iterations, the agent barely reaches heuristic Elo (1585 vs 1584). The heuristic itself is very simple (1-ply lookahead + center preference). A strong Connect 4 agent should significantly exceed this.

Bottleneck diagnosis: The slow Elo growth in Phase 2 (iter 200–2400) suggests the agent gets stuck after learning basic tactics. Potential causes include:
1. **Low entropy** — the policy becomes too deterministic too early, preventing exploration of new strategies
2. **Uniform opponent sampling** — playing against many weak early snapshots provides poor learning signal
3. **Constant hyperparameters** — lr, entropy coef, and clip epsilon may need scheduling to enable different learning phases

Recommended next steps:
1. Use Elo as the primary evaluation metric going forward
2. Investigate targeted algorithmic changes one at a time, using Elo to measure impact
3. Specifically test: higher entropy coefficient, entropy annealing, weighted-recent opponent sampling, and learning rate scheduling